AP PRECALCULUS

Unit 1 • Lesson 1.13 notes

Lesson 1.13 • Function Model Selection and Assumption Articulation

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

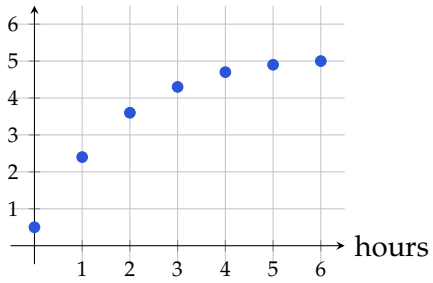
- Choose a function type that fits data or a context.
- State the assumptions behind a model and restrict its domain and range.

Vocabulary. model, linear, quadratic, polynomial, rational, piecewise, assumption, restricted domain.

1. Matching patterns to function types

Pattern in the data or context	Model
roughly constant rate of change	linear
constant second differences; one maximum or minimum	quadratic
two turning points, or a change in concavity	cubic
approaches a limiting value; "per unit" quantities	rational
different rules on different intervals (rates, tax brackets)	piecewise

Example 1 • Scatter plot



The concentration of a medicine rises quickly, then levels off near 5.

Solution. The data approach a limiting value, so a rational model such as $C(t) = \frac{5t+a}{t+b}$ (horizontal asymptote $C = 5$) fits better than a polynomial, which would eventually grow without bound.

2. Assumptions and domain

Every model assumes the observed pattern continues. Restrict the domain to inputs that make sense (time ≥ 0, lengths positive) and to the range of the data when predictions far outside it would be unreliable.

Example 2 • Piecewise

A parking garage charges \$3 for the first hour and \$2 for each additional hour or part of an hour, with a daily maximum of \$15.

Solution. For $0 < t \leq 1$: \$3. Then \$2 per extra hour (a step function) until the cost reaches \$15 at $t = 7$; after that the cost stays \$15. A piecewise model with a cap is needed.

Common mistakes

- Choosing a model only because it fits the plotted points, without checking that it behaves sensibly outside them.
- Using a model for inputs outside its domain (for example, negative times).

Quick check

1. Which model fits "each year the value drops by \$1200"?
2. Why restrict a quadratic population model to the time before it reaches 0?

Answers. 1. Linear. 2. Negative populations are impossible.